MA3521 — Linear ODEs

Name:

Problem 1 $2y' + 2\ln(t)y = 2\exp(t - t\ln(t))$ $\quad y(1) = 0$

1a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\quad q(t) =$

1b. Find an integrating factor.

$$\mu(t) = \exp\left(\int \quad dt\right) =$$

1c. Multiply by $\mu(t)$ and integrate.

$$(\mu y)' =$$

$$\mu(t)y(t) = \int \quad dt + C = \quad + C$$

1d. Use the initial condition to evaluate C.

$C =$

Problem 2 $3y' + 3te^t y = 6t \exp\left((1-t)e^t\right) \qquad y(1) = 2$

2a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\qquad$ $q(t) =$

2b. Find an integrating factor.

$\mu(t) = \exp\left(\int \quad dt\right) =$

2c. Multiply by $\mu(t)$ and integrate.

$(\mu y)' =$

$\mu(t)y(t) = \int \quad dt + C =$ $\qquad + C$

2d. Use the initial condition to evaluate C.

$C =$